

Ryan Peach

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Objective: Senior infrastructure engineer with a decade of experience running large-scale distributed systems on AWS and Kubernetes, including custom controllers for high-performance compute workloads.

EDUCATION

Georgia Tech	Atlanta, GA
<i>MS in Computer Science, Machine Learning Specialty. Computer Vision & Robotics Equiv. Coursework</i>	2015 – 2018
Western Kentucky University	Bowling Green, KY
<i>BS in Electrical Engineering, Minor in Mathematics</i>	2010 – 2014

MAJOR SKILLS

<i>Docker, Kubernetes, AWS</i>	<i>Expert, 2017-Present</i>
<i>CI/CD & DevOps: GitHub Actions, Jenkins, GitOps, Helm</i>	<i>Expert, 2017-Present</i>
<i>Infrastructure as Code: Terraform, CloudFormation</i>	<i>Expert, 2017-Present</i>
<i>Python Scientific Stack, Data Science</i>	<i>Expert, 2013-Present</i>
<i>Machine Learning: Deep Learning, Reinforcement Learning</i>	<i>MS, 2015-Present</i>
<i>Rust Language</i>	<i>Advanced, 2019-Present</i>

EXPERIENCE

Senior Principal Engineer	2023-2026
<i>Ansys / Synopsys</i>	<i>Atlanta, GA</i>

Architecture

- Architected globally available infrastructure spanning multiple regions: global databases, failover clusters, edge services, and read replicas to ensure low-latency access and high availability worldwide.
- Extended Kubernetes platform from cloud-only to hybrid on-prem and cloud environments, enabling deployment flexibility across enterprise customer sites.
- Established enterprise Terraform practices: standardized module structure, remote state management, and onboarding patterns for teams new to IaC.
- Implemented drift detection, policy standards, and Kyverno admission controls across Kubernetes clusters to enforce compliance and security posture at scale.
- Designed and maintained authoritative infrastructure diagrams covering dozens of interoperating services; trained management on enterprise infrastructure model.
- Designed GraphQL, gRPC, and OpenAPI APIs for core backend services, enabling independent service deployments.
- Built a source-of-truth data pipeline extracting state from Kubernetes, landing raw data in S3 for analysis and simultaneously writing to DynamoDB for low-latency operational reads.

Large Language Models

- Participated in early open source efforts with LangChain, LlamaIndex, etc. Contributing code to repos.
- Advised on company LLM projects across multiple teams, attended conferences on LLM research

Director of Cloud DevOps	2021-2023
<i>OnScale</i>	<i>Atlanta, GA</i>

Engineering Leadership

- Organized teams around blast radius boundaries — each team owned a bounded set of services with defined failure domains and on-call scope.
- Led HPC, Infrastructure, and ML teams through the Ansys acquisition, transitioning all to new management upon integration before moving to technical track.

Architecture

- Executed a strangler fig migration from a monolithic API to a microservices architecture, enabling independent deployments and clear team ownership.

- Applied Domain-Driven Design to decompose the platform into bounded contexts; established explicit API contracts between services enforced via OpenAPI specs.
- Authored and maintained C4 architecture diagrams as the authoritative platform reference across the engineering org.
- Managed custom Kubernetes controller orchestrating simulation jobs on high-performance compute (HPC) infrastructure.
- Mandated GitOps workflows org-wide; overhauled CI/CD pipelines to enforce branch protection, automated testing, and deployment gates.
- Stood up full observability stack (metrics, distributed tracing, structured logging) and led FinOps practice — reducing cloud spend while improving incident response.
- Built Plotly dashboards surfacing key platform and business metrics, giving engineering and leadership real-time visibility into system health and usage.

ML Infrastructure

- Architected end-to-end ML pipeline on AWS: ETL ingestion, ECS-based training, and automated periodic retraining — enabling a customer-facing simulation cost estimator.
- Owned cloud resource allocation and cost governance for the ML platform, balancing training throughput against AWS spend.

Machine Learning Researcher - Lead Architect

2017 – 2021

Keysight Technologies

Atlanta, GA

Architecture

- Built a CloudFormation stack using ECS, Lambda that spins up thousands of Docker containers to train, tune, and select optimal SVG models. Weeks of compute completed in an hour regardless of scale.
- Leading the team in code reviews, managed all the DevOps for the team including managing the git repo and the Jenkins CI/CD.
- Designed PC cluster hardware; managed on-prem Unix, Spark, PostgreSQL, MongoDB, and Kubernetes.
- Built ETL pipelines on AWS ingesting data into Redshift via S3 and EMR/Spark; worked with Glue for catalog and job orchestration.

Machine Learning

- Independently investigated almost a dozen potential ML applications at the company, providing preliminary results and feasibility studies.
- Wrote approximately 7 internal publications on patentable inventions, as well as giving multiple company-wide presentations on my work.
- Implemented large scale hyperparameter tuning systems on AWS. Some custom and others using SageMaker. Worked with AutoML solutions.
- Applied Reinforcement Learning to multiple novel problem domains using Monte Carlo Tree Search and Q Learning.
- Data visualization and web-enabled dashboard creation in Tensorboard, Plotly, Dash, Django, Tableau (similar software), Matplotlib, Altair, and Jupyter.

Leadership

- Promoted from Developer to Researcher within only 1 year due to outstanding performance. Won top 10% performer every year with the company.
- Interviewed over a dozen candidates within 3 years.
- Onboarded 3 coworkers in Python best practices and professional software design principles.
- Maintained coding standards, code review, and planning of development milestones.

TECHNICAL SKILLS

Primary Languages: Python, Rust, Bash/Unix, PostgreSQL, Terraform, CloudFormation

Additional Known Languages: Golang, C/C++, C#, Java, Spark/Scala, Cypher

Technologies: Linux, Git, Docker, Kubernetes, ECS, Kyverno, OPA, Supabase, Redis, MongoDB, DynamoDB, Neo4j, GitHub Actions, AWS, Datadog, VIM, Regex